UNIVERSALLY BOUNDED FIBRES BOUND THE CARDINALITY OF TOPOLOGICAL FIBRES

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 426-442, immediately following the statement at 416-424.

Editorial note. One cross-chapter pointer (Properties of Spaces) is recorded in assembly.external_references; the definitional anchor is a same-chapter line-indexed stub.

Allowed standard prerequisites (editorial).

- Definition of universally bounded fibres. Definition definition-universally-bounded of this chapter, cited at proof line 5700 area (proof line 426-428). Supplies the integer n bounding degrees of Spec(k) x_X U over k.
- Fibred products with field points and underlying topological fibres. Properties of Spaces, Lemma spaces-properties-lemma-points-cartesian, cited at proof lines 434-436. Identifies the inverse image of x in |U| with the image of the top horizontal arrow of the fibred-product diagram.

Source reference. Original definition 03JL: sources/stacks-decent-spaces/pinned-original-source.tex, source lines 277–277. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 03H4: focused-seeds/geometry/S9-decent-universally-bounded-finite-fibres/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let S be a scheme. Let X be an algebraic space over S, and let U be a scheme over S. Let $\varphi : U \to X$ be a morphism over S. If the fibres of φ are universally bounded, then there exists an integer n such that each fibre of $|U| \to |X|$ has at most n elements.*

Proof. The integer n of Definition 03JL works. Namely, pick $x \in |X|$. Represent x by a morphism $x : \mathrm{Spec}(k) \to X$. Then we get a commutative diagram

$$\begin{array}{ccc} \mathrm{Spec}(k) \times_X U & \longrightarrow & U \\ \downarrow & & \downarrow \\ \mathrm{Spec}(k) & \xrightarrow{x} & X \end{array}$$

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which shows (via Properties of Spaces, Lemma 03H4) that the inverse image of x in $|U|$ is the image of the top horizontal arrow. Since $\mathrm{Spec}(k) \times_X U$ is finite of degree $\leq n$ over k it has at most n points. $\square$